intake, but with insufficient protein consumption, and this feature distinguishes it from marasmus.

- **Steps to alleviate malnutrition**

Health Promotion:

- Measures direct to pregnant and lactating women (education, supplement)
- Promotion of breastfeeding
- Development of low cost weaning food
- Educating mothers to plan and space child birth
- Ensuring food security in the home
- Ensuring clean environment

Specific protection:

- Increase intake of protein and energy rich food (e.g. milk, egg, fresh fruits)
- Educating people towards regular immunization
- Promote intake of fortified foods.

Early diagnosis and treatment:

- Baseline information
- Early detection of clinical signs of protein energy malnutrition
- Regular assessment of body mass index
- Biochemical and laboratory investigations.

Rehabilitation:

- Hospital treatment
- Nutritional rehabilitation services (NRC i.e. Nutritional Rehabilitation Centers)
- Follow up care

SUMMARY

- Nutrients are essential for providing nourishment to the body. They are broadly classified as macronutrients and micronutrients.
- Humans contract nutritional deficiencies if the nutrients are not consumed in right amounts and right proportion. Macronutrient deficiencies are mainly related to proteins of the body.
- Protein energy malnutrition results in conditions like Marasmus and Kwashiorkor which predominantly affect children in their developmental years.
- Deficiency of Minerals like calcium causes bone related disorders like osteopenia, osteoporosis.
- Iron deficiency anemia can be prevented by consumption of iron rich foods like ragi, dates, jaggery and green leafy vegetables.
- Iodine is an essential mineral to prevent goiter.
- Fat soluble and water soluble vitamins are essential for the normal functioning of the body and their deficiencies lead to multiple deficiency disorders.
- Fat soluble vitamin deficiencies are predominantly related to vision, bone health, child birth and blood clotting.
- Water soluble vitamin deficiencies can lead to diseases like beriberi, pellagra, scurvy, megaloblastic anemia and pernicious anemia.
- Water the universal solvent is also considered as a nutrient
- Water dehydration and intoxication can also lead to fatal consequences.

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